

Local-Currency DeFi Lending under Joint FX and Crypto Shocks: A Path-Dependent Credit-Loss Stress Test

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Abstract

Local-currency depreciation can reduce the USD repayment burden of debt without establishing the safety of a lending protocol. We develop a reproducible counterfactual credit-loss stress test for Argentine-peso (ARS) and Turkish-lira (TRY) lending backed by ETH or BTC. Empirical market inputs comprise 42 joint monthly returns, constructed from official monthly-average exchange rates and monthly averages of daily crypto reference prices over January 2020–July 2023. The initial collateral distribution, auction frictions, rate policies and funding architecture are illustrative assumptions, not observed local-currency lending conditions. Under a three-month moving-block bootstrap, the baseline adaptive rule and timely auctions produce expected liquidated-exposure shares of 41.41% for ARS/ETH and 42.11% for TRY/ETH. Mean credit losses are 2.00% and 1.46% of initial principal; 99% conditional value-at-risk (CVaR) is 19.18% and 15.81%. Close-out extensions prevent delayed liquidations from disappearing at the horizon. Principal-weight ablations show that MakerDAO draw sizes serve a scale illustration rather than percentage-loss calibration. Within the tested policy grid, rate pass-through trades off financing-side debt relief against liquidation and credit loss. Results depend materially on monthly measurement, the short calibration sample and auction assumptions. Credit-budget exhaustion, a stylized matched-funding accounting deficit and stablecoin redemption failure are distinct outcomes; this study does not forecast peg stability or establish protocol viability.

Keywords: DeFi; local-currency debt; credit loss; liquidation; foreign exchange; counterfactual stress testing.

1 Introduction

Collateralised decentralized-finance lending combines a loan claim, pledged crypto collateral and a token or funding obligation. Smart-contract accounting makes these links explicit, but does not eliminate credit, liquidity or governance risks [1, 2, 12]. A local-currency unit adds an important distinction: depreciation against USD reduces the translated value of both an LCU loan and a matching LCU liability. It is therefore incorrect to infer a protocol loss directly from a borrower’s USD debt erosion.

We study a hypothetical architecture in which the protocol advances one LCU loan funded by an equal, non-interest-bearing LCU token liability. One token promises one LCU, not one USD. The loan accrues interest, while the original funding principal does not. Crypto collateral remains pledged borrower property until sale; it must not also be counted as an unencumbered protocol

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asset. USD reserves supplied by equity investors absorb a settlement deficit. If those reserves are inadequate, the model reports an unfunded accounting gap; it does not assume successful governance-token issuance or peg restoration. Redemption is assumed to coincide with loan settlement at the model FX rate. On-demand redemption, runs and executable exchange-market access are outside this stylized funding diagnostic.

Primary estimand and unit. The primary estimand is expected cumulative contractual credit loss after collateral recovery, divided by initial USD principal, for a synthetic currency–collateral portfolio over a 12-month origination horizon plus any scheduled liquidation close-out. A unit of simulation is one joint market path applied to an assumed distribution of initial collateral ratios (CRs), not an observed MakerDAO borrower.

The questions are how lagged rate policies alter debt growth and liquidation, how auction frictions and collateral buffers change losses, and how much these outcomes vary with market measurement and calibration. The contribution is an integrated, reproducible accounting experiment. It combines monthly first-breach rules with explicit delayed settlement, a proceeds waterfall, separate survivor and liquidated-borrower measures, and matched-liability diagnostics. It makes no universal impossibility claim and provides no regulatory capital recommendation.

2 Related literature and incremental contribution

Empirical DeFi lending studies establish that rates, utilization and borrower activity are state-dependent [3–5]. Liquidation studies show that collateral thresholds and execution incentives matter, and that observed liquidations interact with market liquidity and price volatility [6–8]. Evidence on MakerDAO resilience and oracle design motivates attention to collateral valuation and execution [9, 10]. These results justify modelling frictions; they do not identify the numerical haircut or congestion coefficient for a hypothetical ARS or TRY protocol.

Stablecoin research distinguishes backing from secondary-market stability: arbitrage, endogenous collateral dynamics and redemption institutions all matter [14–16]. A credit-loss calculation cannot replace a run model. Emerging-market research highlights currency substitution and financial-stability exposures [17, 18]. Evidence connecting stablecoin flows and FX markets further motivates treating official FX as a reference unit rather than a guaranteed conversion price [19].

Table 1 selects representative works for their relevance to a particular model channel, not as an exhaustive review of each field. The novelty is the integration and auditability of those channels under local-currency denomination, not the invention of liquidation stress testing, bootstrap inference or stablecoin balance-sheet accounting.

Table 1: Representative literature and the boundary of this experiment.

Channel	Established contribution	Increment here / omitted mechanism
Lending [3, 5]	Empirical rates, usage and liquidity	Compare specified LCU rate rules; demand is fixed.
Liquidation [6–8]	Threshold exposure and execution frictions	Joint FX–crypto paths with scheduled close-out; no order book.
Stablecoins [14–16]	Arbitrage, backing and redemption fragility	Separate loan recovery from funding; no endogenous peg/run.
Emerging-market FX [17–19]	Currency substitution and market segmentation	Explicit reference FX numeraire; no parallel-rate estimate.

Relationship to the companion paper. The companion working paper [13] decodes MakerDAO ETH-A draws and evaluates fixed-horizon debt erosion. The present study reuses its event eligibility

rules, principal summary and 100 quantile bins, plus the official FX snapshots; these are not new data collection. The basic USD translation identity is also shared. New analysis consists of the joint FX–crypto return panel, monthly rate and collateral paths, liquidation and close-out accounting, funding bridge, CR and auction sensitivity, and sampling diagnostics. Results here are freshly simulated, not the companion paper’s event-level estimates. The 130,742 draws are neither 130,742 independent return observations nor observed LCU loans.

3 Accounting architecture and notation

3.1 Balance sheet and loss definitions

Let E_t be LCU per USD and P_0 initial USD principal. Funding tokens have face value $P_0 E_0$ LCU, so their USD value before retirement is $P_0 E_0 / E_t$. Accrued loan assets are larger by contractual interest. Table 2 prevents double counting of collateral and reserves.

Table 2: Illustrative balance sheet before settlement. Values shown for one loan; no claim is made that this is an implemented protocol.

Item	Native unit	USD treatment and ownership
Loan asset	LCU	Accrued claim $P_0 U_t / X_t$, before credit impairment.
Funding/token liability	LCU	Non-interest-bearing face value P_0 / X_t ; one token redeems for one LCU at settlement.
Pledged crypto	ETH or BTC	$P_0 c_j Z_t$; borrower property securing the loan, not an additional unencumbered asset.
Liquid reserves	USD	$k P_0$ initially, with $k = 0.10$ in funding diagnostics.
Equity	Residual value	USD Reserves plus accrued net interest less impairment; absorbs losses first.
Governance back-stop	None credited	No token-sale proceeds, guarantees or recapitalization are assumed.

Five concepts must be separated. (i) Contractual credit loss is debt minus loan recovery. (ii) FX translation changes the USD loan value. (iii) Matching token liabilities translate with the same FX factor, offsetting the principal component. (iv) Peg or redemption shortfall also depends on timing, liquidity and market conversion; it is not estimated here. (v) Overall protocol insolvency would require all assets and obligations, valuation rules and contingent liabilities. We therefore use *credit loss* throughout and reserve the term *accounting gap* for the deliberately restricted funding ledger.

3.2 Variables and units

Table 3: Notation. All lower-case monetary outcomes are normalized by their own initial USD principal; portfolio weights sum to one.

Symbol	Definition / unit
s, j, b, t	Market path, collateral cohort, principal bin, and month indices.
N, T, d, τ_j	Number of paths; $T = 12$ origination months; execution delay in months; first breach month.
E_t, P_t, X_t, Z_t	LCU/USD FX level; USD/crypto price; factors $X_t = E_t/E_0$, $Z_t = P_t/P_0^{\text{crypto}}$.
δ_t, g_t, r_t, U_t	FX depreciation return; crypto return; nominal annual borrowing rate; cumulative LCU debt factor.
c_j, w_j, θ	Initial CR; initial-principal exposure weight; liquidation CR threshold.
$q_t, h_0, \gamma, h_{s,t}, \bar{h}$	Initial-exposure share sold in month t ; base haircut; congestion slope; effective haircut; haircut cap.
D_{sjt}, C_{sjt}, a_{sj}	Debt and collateral in USD per initial USD; auction proceeds at settlement.
$R_{sj}, L_{sj}, \pi_{sj}, V_{sj}$	Debt recovery, credit loss, collectible penalty, and returned collateral value per initial USD.
$e_{sj}, b_{sj}, k, K_{s,t}$	Settlement month; financing-side net debt effect; initial reserve ratio; settlement capital ledger per initial portfolio USD.
$\ell_s, \alpha, \text{CVaR}_{.99}$	Portfolio loss ratio; tail probability $\alpha = .01$; mean of worst 1% of simulated losses.

3.3 Debt and liquidation equations

For monthly returns δ_t and g_t , and a nominal annual rate r_t , the explicit recursions are

$$X_t = X_{t-1}(1 + \delta_t), \quad Z_t = Z_{t-1}(1 + g_t), \quad (1)$$

$$U_t = U_{t-1}(1 + r_t/12), \quad U_0 = X_0 = Z_0 = 1, \quad (2)$$

$$D_{sjt} = U_{s,t}/X_{s,t}, \quad C_{sjt} = c_j Z_{s,t}, \quad CR_{sjt} = C_{sjt}/D_{sjt}. \quad (3)$$

The rate convention is nominal annual divided by 12, not effective annual compounding. First breach uses the strict inequality $CR < \theta$. A breach in months 1–12 is irrevocable and schedules full collateral sale in month $\tau_j + d$, even if the CR later recovers. Interest accrues until execution. Unbreached positions repay in month 12. No new loans or breaches are originated after month 12; months 13–14 only close scheduled sales.

For exposure sold in month t , $q_{s,t} = \sum_j w_j \mathbf{1}(e_{sj} = t, \tau_j \leq 12)$, and

$$h_{s,t} = \min(\bar{h}, h_0 + \gamma q_{s,t}), \quad \bar{h} = 0.30, \quad (4)$$

$$a_{sj} = (1 - h_{s,e_{sj}}) C_{sj,e_{sj}}, \quad (5)$$

$$R_{sj} = \min(D_{sj,e_{sj}}, a_{sj}), \quad L_{sj} = D_{sj,e_{sj}} - R_{sj}, \quad (6)$$

$$\pi_{sj} = \min(0.13 D_{sj,e_{sj}}, [a_{sj} - D_{sj,e_{sj}}]_+), \quad (7)$$

$$V_{sj} = [a_{sj} - D_{sj,e_{sj}} - \pi_{sj}]_+. \quad (8)$$

Here $[x]_+ = \max(x, 0)$. Penalty is paid to the protocol only from proceeds remaining after loan recovery; it cannot increase loan recovery above debt or create a fictitious cash receipt. The auction haircut represents value lost to execution frictions or auction participants. For survivors, $R = D$, $L = \pi = h = 0$, and collateral remains with the borrower. The liquidation waterfall satisfies $a = R + \pi + V$ exactly.

The financing-side net debt effect is

$$b_{sj} = 1 - (R_{sj} + \pi_{sj} + h_{s,e_{sj}} C_{sj,e_{sj}}). \quad (9)$$

For liquidation this equals $1 - C + V$; for survival it is $1 - D_{sj,12}$. It measures normalized initial financing proceeds less resources surrendered in settlement, including auction friction. It excludes pre-sale collateral investment returns, taxes, conversion costs, peg deviations, discounting, post-settlement reinvestment and borrowing utility. Default can reduce repayment, so it is not a welfare, profit or general borrower-benefit measure. Liquidated and surviving positions have different holding periods; we report both separately.

3.4 Funding and reserve diagnostics

Gross portfolio credit loss is $\ell_s = \sum_j w_j L_{sj}$. Its credit-budget indicator is $\mathbf{1}(\ell_s > k)$, not an insolvency event. In the separate matched-funding diagnostic, outstanding principal liabilities are retired at each position's settlement FX rate. The cash-capital ledger is

$$K_{s,t} = k + \sum_{j:e_{sj} \leq t} w_j \left(R_{sj} + \pi_{sj} - X_{s,e_{sj}}^{-1} \right). \quad (10)$$

Equivalently, each settled contribution is $(U - 1)/X - L + \pi$. Thus the loan and liability principal translation cancels; interest is not assumed to offset loss until actually recovered. A negative ledger flags a funding gap, even if it later recovers. Negative balances are recorded as unfunded gaps, not assumed external borrowing. Before all loans settle, this cash ledger is not total marked-to-market equity; unrealized loan surplus is excluded. After all settlement it equals residual equity for this restricted architecture. USD conversion at official reference FX and synchronized liability maturity are strong assumptions. Early token redemption, liquidity runs, operational costs and endogenous token prices are not simulated.

4 Data, construction and parameter provenance

4.1 Market observations and alignment

Archived OECD/FRED ARS and TRY monthly-average series provide LCU per USD [20, 21]. Coin Metrics daily `PriceUSD` observations are averaged within each calendar month [22]. There are 43 level observations, January 2020–July 2023, yielding 42 returns. This revision replaces the original mixture of monthly-average FX and month-end crypto. Both main series now use monthly averages, although business-day FX and calendar-day crypto sampling are not identical. These are low-frequency reference marks, not simultaneous executable prices; monthly averaging also suppresses intramonth extremes.

Official rates may differ substantially from accessible conversion prices, especially under Argentine exchange controls. No parallel-rate observations are invented or substituted. This is an official-rate counterfactual, and its conclusions should not be interpreted as returns achievable by an Argentine retail borrower. The archived analysis period is held fixed for traceability. Subperiods and leave-one-year-out analyses test sample dependence; a longer modern sample remains a distinct future data extension.

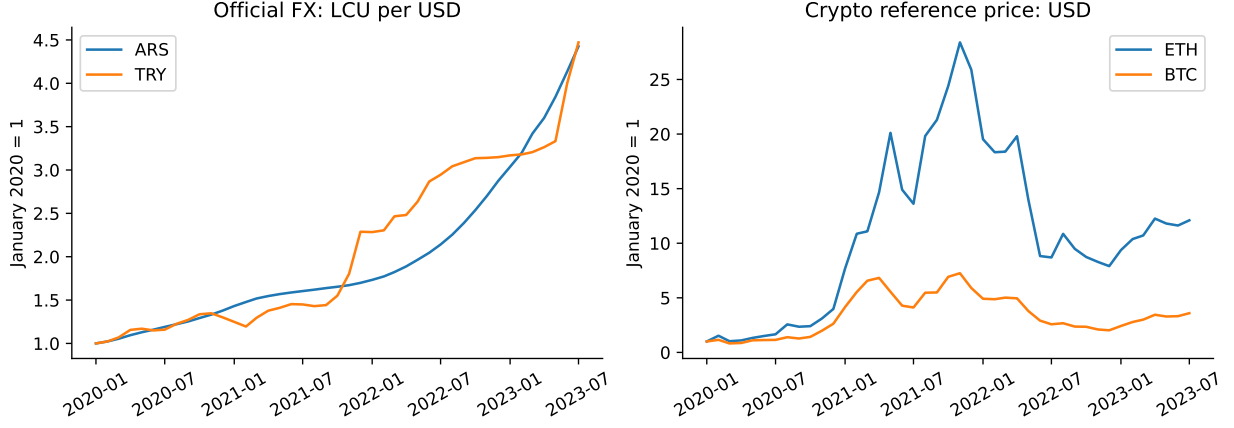


Figure 1: Monthly-average FX and crypto price factors. LCU means local-currency units. Each series is normalized to its January 2020 level; these are observed reference prices, not simulated loan outcomes.

4.2 MakerDAO principal ablation

The companion package derives 130,742 eligible ETH-A debt draws from a public MakerDAO dataset [11, 13]. This revision uses its archived 100 principal-bin summaries as an inherited scale and concentration illustration. Raw blockchain decoding was not repeated for this revision. The input-preparation script can rebuild those summaries from the documented companion event file and verifies its checksum when supplied.

Let v_b be the principal share of bin b and w_j the assumed CR exposure weight. The synthetic product portfolio has weights $v_b w_j$ across $100 \times 31 = 3,100$ cells. Principal size is assumed independent of CR and market path; all dollar amounts scale linearly and congestion depends on a share. Therefore for any normalized cohort outcome f_j ,

$$\sum_b \sum_j v_b w_j f_j = \sum_j w_j f_j, \quad \sum_b v_b = 1. \quad (11)$$

This is exact scale homogeneity under the stated model, not empirical evidence that size never matters. Dollar-dependent slippage or correlated size and leverage would break it.

Table 4 runs MakerDAO weights, equal principal bins and one normalized unit per CR quadrature point retaining w_j . It additionally implements the literal alternative of one dollar at each of the 31 CR points. The latter makes the CR distribution uniform, so any percentage change is a collateral-weight change, not an effect of MakerDAO draw sizes.

Table 4: Principal ablation: ETH, adaptive rule, timely auction, common paths. Liquidated exposure, mean credit loss and CVaR₉₉ are percentages of initial principal.

Currency	Scheme	Cells	Liquidated	Loss	CVaR ₉₉
ARS	MakerDAO weights	3100	41.41	2.00	19.18
ARS	Equal principal bins	3100	41.41	2.00	19.18
ARS	Normalized CR quadrature	31	41.41	2.00	19.18
ARS	One dollar per CR point	31	40.69	1.70	14.08
TRY	MakerDAO weights	3100	42.11	1.46	15.81
TRY	Equal principal bins	3100	42.11	1.46	15.81
TRY	Normalized CR quadrature	31	42.11	1.46	15.81
TRY	One dollar per CR point	31	41.63	1.23	11.34

4.3 Parameters and provenance

No auction parameter is presented as empirically estimated. Four provenance classes distinguish observations, external methodological inputs, inherited protocol conventions, and illustrative numerical choices. Table 5 contains the full numerical policy specification; constants also appear in executable code.

Table 5: Parameter provenance. “Inherited convention” means a design pattern, not a verified deployment-specific parameter.

Class	Input / value	Interpretation and source
Directly observed	43 monthly levels; 42 returns	Archived FRED FX and Coin Metrics daily prices, aggregated to monthly averages.
External source	Block resampling; Maker-DAO bin summaries	Dependence-respecting resampling [23]; inherited draw-size summary [11, 13], not an LCU calibration.
Inherited protocol convention	Collateralized debt, accrued rates, liquidation accounting	Motivated by Maker/Sky core accounting [12]; this implementation is not a reproduction of its contracts.
Illustrative	CR 1.50–2.25, 31 points; triangular mode 1.75	Endpoints assigned positive mass 10^{-6} before normalization. Alternatives: uniform; density $\exp[-8(c - 1.5)]$; uniform 2–3 buffered grid.
Illustrative	Threshold 1.50; penalty 0.13	Carried over as design assumptions; no historical calibration claimed.
Illustrative	Haircut 0.08; slope 0.20; cap 0.30	Joint grid: base haircut 0.05/0.08/0.15, slope 0/0.20/0.40, delay 0/1/2 months, all four CR distributions.
Illustrative	Reserve 0.10; 12-month horizon	Zero reserve yield; liability principal retires at loan settlement. Extension up to two months.
Illustrative	Rate floor 0.05; initial rate 0.20	Coefficients, caps and adjustment rules are specified below; not estimated optimal policies.
Numerical choice	20,000 main paths; block length 3	Seed 20260928 with deterministic currency/asset offsets; 5,000 common paths for each joint-grid cell; 100 outer replicates, 2,000 inner paths.

5 Stress-test design and event order

The main bootstrap samples joint FX–crypto pairs in non-circular blocks of three consecutive observations. Starting indices are uniform over $0, \dots, n - B$; blocks that extend beyond the historical sample are never eligible. Blocks are concatenated and the final partial block is truncated to 14 months. The same 14-month paths are used for all rate and auction comparisons within a currency–collateral pair. No path extension is deterministically extrapolated. Alternative block lengths 1 and 6 use their own fixed seeds; the block-3 robustness row reuses the main run exactly.

Each month: (1) observe reference FX and crypto prices; (2) determine rates using only prior FX returns; (3) accrue unpaid debt; (4) calculate CRs; (5) register first strict threshold breaches through month 12; (6) queue them for the specified execution delay; (7) execute due full-collateral sales and determine congestion; (8) allocate proceeds to debt, penalty and borrower residual, and record loss; (9) retire matching funding, update the capital ledger and borrower outcomes. Unbreached loans repay at month 12; scheduled sales continue through month 14. A standalone one-page algorithm accompanies the revision.

The historical diagnostic uses every complete 14-month market window: 29 overlapping windows per currency–collateral pair. Each provides 12 months for origination and two for possible close-out. A common available-window set avoids comparing timely full windows with shorter delayed samples. These windows overlap and are not independent observations.

Twenty thousand simulated paths reduce numerical integration error conditional on 42 monthly observations. They do not create additional economic observations. We add calendar-aware leave-one-year-out analysis (blocks cannot cross a deleted calendar year), early and late subperiods, an IID bivariate Gaussian log-return benchmark with sample-estimated mean and covariance, and outer block resampling of the 42-return calibration. The Gaussian benchmark preserves contemporaneous covariance but not serial dependence or fat tails; it is a model comparison, not a preferred forecast. Outer-resampling percentile ranges combine calibration perturbations with residual inner simulation noise. With only 100 outer replicates they are descriptive sensitivity bands, not coverage-validated confidence intervals. Joint parameter ranges represent scenario uncertainty rather than probabilities over parameters.

6 Rate rules and diagnostics

For $m_t = \min(k, t - 1)$ prior returns and $t > 1$, define

$$A_{t,k} = \max \left\{ \exp \left[\frac{12}{m_t} \sum_{u=t-m_t}^{t-1} \log(1 + \delta_u) \right] - 1, 0 \right\}. \quad (12)$$

All rules start at $r_1 = 0.20$ and use expanding history until their full lookback is available; no preceding unobserved return is fabricated. Policies are static 20%; adaptive $\text{clip}(0.10 + 0.75A_{t,3}, 0.05, 1.50)$; indexed $\text{clip}(0.05 + A_{t,3}, 0.05, 2.00)$; lower-cap adaptive with a 0.60 ceiling; and trailing-12 adaptive with $k = 12$ and a 0.60 ceiling. Partial adjustment uses the lower-cap target r_t^* and

$$r_t = r_{t-1} + \text{clip}\{0.25(r_t^* - r_{t-1}), -0.05, 0.05\}. \quad (13)$$

Thus governance moves at most five percentage points per month. The trailing-12 policy has fewer than 12 past observations within the initial origination year; full history becomes available only in close-out. We label that initialization explicitly rather than implying a pre-existing 12-month history.

Tables 6–7 report rates actually charged while ETH loan debt remains outstanding under timely auctions in months 1–12, including the standardized initialization month. Statistics weight observations by outstanding initial-principal exposure, not by accrued debt. Rates are nominal annual percentages; cap frequency is a percentage of those exposure-months; change is mean absolute month-to-month movement in percentage points. Full median-change diagnostics are in the CSV. Rates after a cohort settles are excluded; unweighted posted-rate means and medians are also retained in the CSV. We do not add a purported forward-looking rate without forecasts or forward-market data.

Table 6: ARS rate diagnostics. P90/P99 are distribution percentiles; all policy coefficients are assumptions.

Rule	Mean	Median	P90	P99	Max	At cap	Change
Static 20	20.00	20.00	20.00	20.00	20.00	0.00	0.00
Adaptive 75	47.35	41.49	86.05	106.32	109.72	0.00	10.84
Indexed 100	54.73	46.51	105.91	133.43	137.96	0.00	14.32
Trailing 12	42.78	43.49	60.00	60.00	60.00	24.16	4.37
Partial adjustment	34.05	33.82	48.75	55.30	57.33	0.00	2.95
Lower cap	41.88	41.74	60.00	60.00	60.00	27.61	6.63

Table 7: TRY rate diagnostics, with the same units and exposure-month denominator as Table 6.

Rule	Mean	Median	P90	P99	Max	At cap	Change
Static 20	20.00	20.00	20.00	20.00	20.00	0.00	0.00
Adaptive 75	46.41	30.30	124.96	150.00	150.00	7.41	19.74
Indexed 100	53.84	31.14	158.28	200.00	200.00	6.98	26.60
Trailing 12	35.20	32.12	60.00	60.00	60.00	23.26	6.18
Partial adjustment	27.93	25.89	42.62	52.78	57.33	0.00	3.23
Lower cap	34.26	31.01	60.00	60.00	60.00	25.27	10.32

7 Results and denominators

7.1 Main outcomes

Liquidated exposure is $\mathbb{E}[\sum_j w_j \mathbf{1}(\tau_j \leq 12)]$, including all scheduled sales after close-out. Mean credit loss is $\mathbb{E}[\ell_s]$ relative to initial principal. CVaR_{99} is the average of the largest 1% of path-level losses (200 observations in a 20,000-path run). For non-multiples of 100, the implementation uses the largest $\lceil .01N \rceil$ losses. The probability of *any* portfolio liquidation is a different statistic and can be large because a very small positive cohort weight is close to the threshold. It is not an individual borrower’s liquidation probability. Cohort-specific probabilities are provided in `cohort_results.csv`; averaging them with w_j gives the probability for a representative normalized-principal position drawn from the assumed CR distribution, not an empirically identified MakerDAO position.

Table 8 reports six rate rules under timely auctions. The baseline adaptive mean credit loss is 2.00% for ARS/ETH and 1.46% for TRY/ETH, with tail loss 19.18% and 15.81%. Higher pass-through increases liquidation in these cases, while partial adjustment attenuates that response. These are comparisons within the tested design space. A three-way tension among financing relief, rate indexation and low credit loss is not proof that every possible protocol must face the same frontier.

Table 8: ETH, timely execution. All entries other than labels are percentages: Liq. = initial exposure liquidated; Loss = credit loss / initial principal; CVaR_{99} = tail loss / initial principal; Net effect = financing-side effect / initial principal.

Currency	Rule	Liq.	Loss	CVaR_{99}	Net effect
ARS	Static 20	33.64	1.64	14.90	5.64
ARS	Adaptive 75	41.41	2.00	19.18	-15.12
ARS	Indexed 100	43.96	2.18	21.47	-21.48
ARS	Trailing 12	39.26	1.93	17.95	-11.22
ARS	Partial adjustment	36.12	1.80	16.21	-3.96
ARS	Lower cap	39.05	1.91	17.96	-10.41
TRY	Static 20	34.48	1.07	11.52	2.44
TRY	Adaptive 75	42.11	1.46	15.81	-16.33
TRY	Indexed 100	44.87	1.73	17.69	-22.36
TRY	Trailing 12	38.47	1.24	13.27	-8.15
TRY	Partial adjustment	36.13	1.14	12.24	-2.70
TRY	Lower cap	38.24	1.24	13.29	-7.46

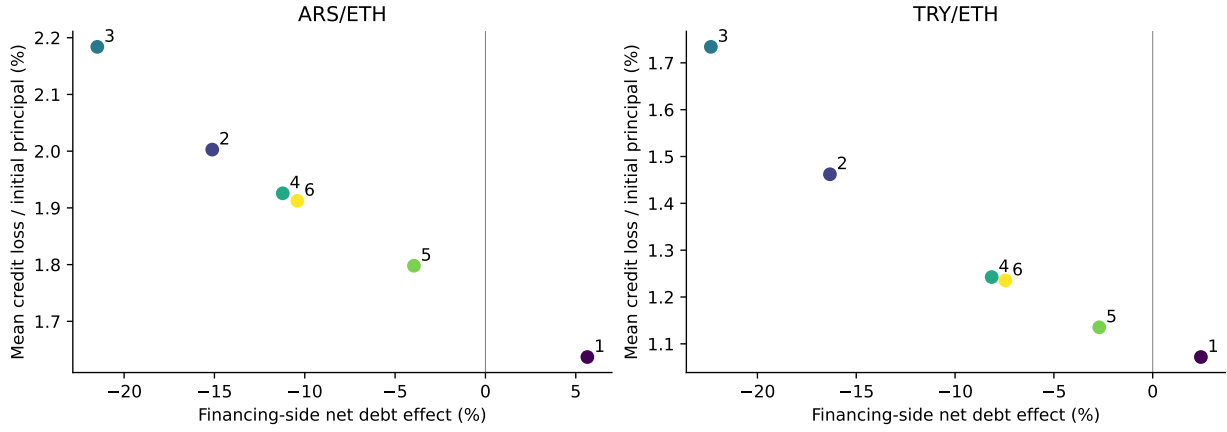


Figure 2: Conditional policy comparison for ETH and timely auctions. Labels 1–6 denote static, adaptive, indexed, trailing-12, partial-adjustment and lower-cap rules. Both axes are normalized by initial principal. This discrete set is not an efficient frontier or impossibility theorem.

7.2 Breach, execution and funding

The horizon treatment makes breach incidence comparable across delay-only and timely scenarios: the underlying breach criterion and path are unchanged. Table 9 distinguishes sales executed by month 12 from pending exposures and reports all losses after scheduled close-out. Haircut-only and delay-only comparisons isolate the separate effects of these frictions. Delay need not increase loss on every path, since collateral may recover before irrevocable execution.

Table 9: ETH, adaptive rule. Breach, executed and pending are percentages of initial exposure; executed + pending = breach at month 12. Loss and CVaR₉₉ include all scheduled close-outs and use initial principal as denominator.

Currency	Auction	Breach	Executed	Pending	Loss	CVaR ₉₉
ARS	Timely	41.41	41.41	0.00	2.00	19.18
ARS	Delay only	41.41	40.67	0.75	4.63	39.66
ARS	Haircut only	41.41	41.41	0.00	3.19	22.57
ARS	Delayed stressed	41.41	40.67	0.75	5.99	42.87
TRY	Timely	42.11	42.11	0.00	1.46	15.81
TRY	Delay only	42.11	41.30	0.81	4.22	34.36
TRY	Haircut only	42.11	42.11	0.00	2.55	19.29
TRY	Delayed stressed	42.11	41.30	0.81	5.59	38.06

Table 10: Different risk measures, ETH adaptive rule. Any liq. = percentage of paths with any cohort liquidation; Under-coll. = initial exposure pending and below CR 1 at month 12; Credit budget = percentage of paths with gross credit loss above 10% of initial principal; Capital gap = percentage with a negative matched-funding settlement ledger.

Currency	Auction	Any liq.	Under-coll.	Credit budget	Capital gap
ARS	Timely	60.30	0.00	8.05	2.62
ARS	Delay only	60.30	0.03	17.44	6.04
ARS	Haircut only	60.30	0.00	10.46	5.00
ARS	Delayed stressed	60.30	0.03	23.07	12.22
TRY	Timely	62.77	0.00	5.15	0.00
TRY	Delay only	62.77	0.01	16.41	5.75
TRY	Haircut only	62.77	0.00	8.82	2.38
TRY	Delayed stressed	62.77	0.01	24.13	8.87

The timely 10% credit-budget breach probabilities are 8.05% and 5.15%. They are neither deposit-redemption probabilities nor general insolvency rates. The funding diagnostic can be smaller because interest receipts and matching LCU principal liabilities matter. Conversely, its synchronized maturity and frictionless reference-FX conversion omit important sources of real funding stress. Zero simulated accounting gaps in a scenario do not establish safety.

7.3 Borrower outcomes

Survivor and liquidated effects in Table 11 divide pooled effects by their respective pooled initial exposure across paths. The all-portfolio effect divides by complete ex ante exposure. These conditional quantities reconcile exactly using the pooled survival weight. Loss/liquidated debt divides aggregate contractual loss by aggregate accrued USD debt of liquidated positions; it is not the mean of pathwise ratios. The measure includes financing-side outcomes of liquidated positions instead of assigning them only a penalty.

Table 11: ETH, timely auctions. Financing-side effects (survivors, liquidated, all) are percentages of the respective initial exposure; the final column is credit loss as a percentage of accrued liquidated debt.

Currency	Rule	Survivors	Liquidated	All	Loss/liq. debt
ARS	Static 20	19.40	-21.48	5.64	5.14
ARS	Adaptive 75	-5.37	-28.92	-15.12	4.81
ARS	Indexed 100	-13.02	-32.27	-21.48	4.82
ARS	Trailing 12	-1.33	-26.52	-11.22	4.97
ARS	Partial adjustment	6.84	-23.05	-3.96	5.18
ARS	Lower cap	-0.26	-26.26	-10.41	4.97
TRY	Static 20	15.12	-21.66	2.44	3.32
TRY	Adaptive 75	-8.03	-27.74	-16.33	3.52
TRY	Indexed 100	-15.19	-31.15	-22.36	3.81
TRY	Trailing 12	2.15	-24.61	-8.15	3.36
TRY	Partial adjustment	8.48	-22.47	-2.70	3.33
TRY	Lower cap	3.06	-24.43	-7.46	3.37

8 Robustness and uncertainty

8.1 Market measurement and short history

Using the mixed monthly convention with the simulation model produces mean losses of 3.43% and 3.16%, versus 2.00% and 1.46% with monthly averages for both series. Measurement choice is therefore material. Appendix tables report block lengths, omitted years, subperiods and the parametric benchmark. Omitting 2022 sharply reduces losses in both currencies; this short history is economically consequential. The parametric model can yield different tail estimates despite fitting first and second log-return moments.

Table 12: ETH adaptive/timely uncertainty. All loss entries are percentages of initial principal. MC halfwidth is 1.96 times the conditional simulation standard error. Input low/high are 2.5/97.5 percentiles of 100 outer-resampled mean losses; tail low/high are the analogous CVaR₉₉ percentiles. They are sensitivity bands, not validated confidence intervals.

Currency	Mean loss	MC halfwidth	Input low	Input high	Tail low	Tail high
ARS	2.00	0.06	0.08	4.44	3.99	20.73
TRY	1.46	0.04	0.28	3.79	6.20	19.49

8.2 Joint auction and collateral assumptions

The joint grid contains $3 \times 3 \times 3 \times 4 = 108$ parameter combinations per currency for ETH and the adaptive rule. Every cell uses the same first 5,000 baseline paths. This isolates parameter changes, although 99% tail estimates then average only 50 paths. Table 13 and Figures 3–4 show the full grid. Loss magnitudes are assumption-dependent; a claim of robust low loss across all plausible designs would be unsupported. Larger collateral buffers generally reduce exposure to liquidation, but auction frictions and delay still matter. The grid does not assign probability or empirical plausibility weights to its cells.

Table 13: Joint sensitivity ranges, ETH adaptive rule. Min/max span haircut, congestion and delay combinations within each CR distribution. All values are loss percentages of initial principal; tail refers to CVaR₉₉.

Currency	CR distribution	Min loss	Max loss	Min tail	Max tail
ARS	base	0.12	6.20	2.73	53.11
ARS	buffered	0.10	2.64	1.88	48.31
ARS	near_threshold	0.16	8.32	3.89	55.53
ARS	uniform	0.16	5.82	2.10	52.03
TRY	base	0.06	6.15	1.97	52.00
TRY	buffered	0.04	2.77	1.07	48.60
TRY	near_threshold	0.06	8.51	1.64	56.04
TRY	uniform	0.07	5.85	1.53	52.43

9 Design implications and limitations

The demonstrated implications are narrow: reporting a gross credit budget as “protocol solvency” obscures funding assumptions; testing auction haircut independently of delay improves interpretation; rate caps and partial adjustment change debt growth; and a short historical calibration produces substantial input uncertainty. An initial-principal ceiling derived solely from CVaR is

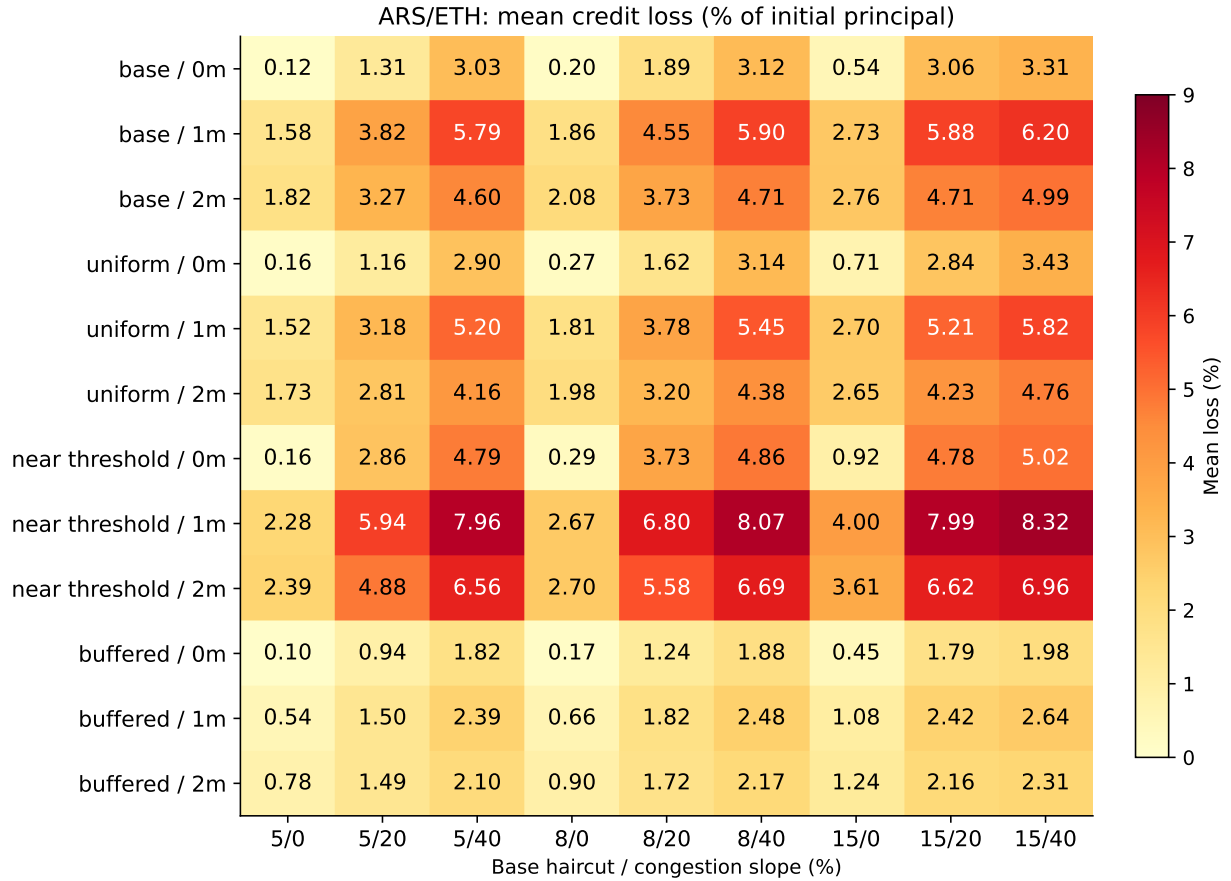


Figure 3: ARS/ETH joint mean-loss surface. Rows specify assumed collateral distribution and execution delay in months; columns give base haircut / congestion slope in percent. Cell values are mean credit loss as a percentage of initial principal. The effective haircut is capped at 30%.

not labelled “safe” here. Diversified auction participants, circuit breakers and backstop liquidity could change the assumed haircut or execution mechanism, but their efficacy is not tested and requires separate investigation.

Several constraints remain. Monthly-average reference marks miss intramonth triggers and executable price paths. Official FX may be inaccessible. The 42-month calibration contains unusual economic regimes, and block resampling assumes enough local stationarity to reuse them. The parameter grid is a design exercise, not an estimated population distribution. Principal and leverage are independent by assumption, while real borrowers can adjust collateral, repay, refinance or default strategically. Auctions sell the entire position, do not simulate order books or induced price feedback, and impose an illustrative linear congestion rule. The funding diagnostic excludes early redemption, endogenous stablecoin prices, legal priority, operational costs, cross-currency hedges and governance recapitalization. Longer data, an executable FX dataset and a separately specified redemption/run model would be required for stronger claims.

Data alignment, initialization, close-out censoring, cohort alternatives and parameter sensitivity have been addressed directly rather than left as disclaimers. This does not remove their residual modelling uncertainty.

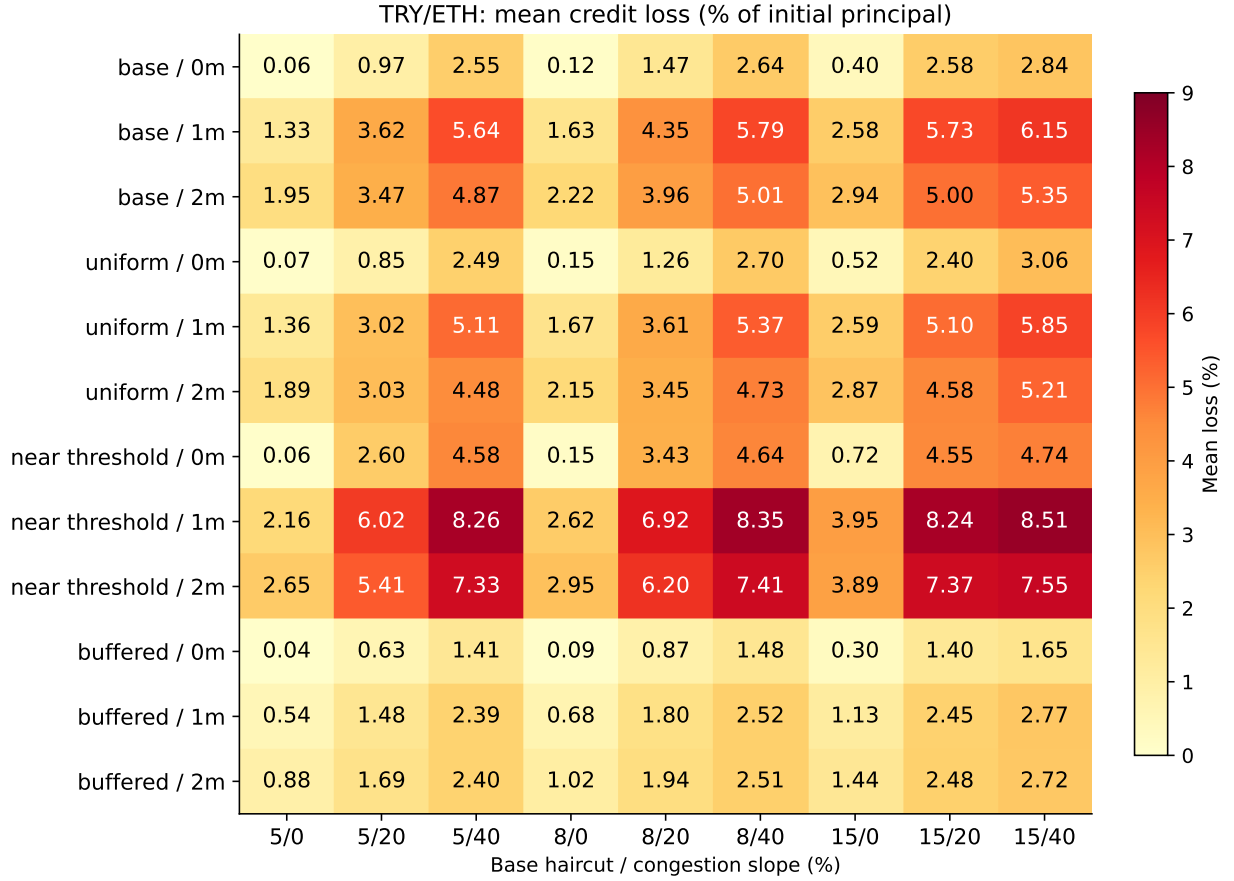


Figure 4: TRY/ETH joint mean-loss surface, using the same units, grid, rate rule and common-path design as Figure 3.

10 Conclusion

This study provides a transparent counterfactual credit-loss framework linking local-currency debt growth, crypto collateral and liquidation settlement. On the stated monthly-average calibration, the adaptive timely-auction mean losses are 2.00% and 1.46% for ARS/ETH and TRY/ETH. Principal-weight invariance clarifies the descriptive role of MakerDAO draw sizes, while close-out accounting distinguishes pending sales from executed liquidations. Measurement, economic subperiod and auction assumptions materially affect results. The contribution is a reproducible stress-testing method and an explicit accounting boundary, not evidence that an actual local-currency lending protocol is viable, solvent or safe.

Reproducibility and revision materials

The repository supplies raw market extracts, transformation code, model tests, machine-generated CSVs, figures, parameter provenance, a one-page algorithm, a point-by-point response and a headline reconciliation table. `bash run_all.sh` rebuilds inputs, simulations, exhibits and PDFs. A frozen checksum manifest, exhibit-to-source map and `analysis/verify_reproduction.py` permit file-level comparisons and independent reconstruction of the monthly panel. The event-level MakerDAO CSV is linked at a pinned companion-repository commit and can be checked and used to rebuild the inherited principal bins; raw blockchain decoding is not repeated here. Those bins do not affect main percentage outcomes.

A Additional generated results

Table 14: BTC, timely auctions. Same definitions and percentage denominators as Table 8.

Currency	Rule	Liq.	Loss	CVaR ₉₉	Net effect
ARS	Static 20	42.30	0.32	5.13	0.62
ARS	Adaptive 75	52.25	0.49	7.68	-20.13
ARS	Indexed 100	55.13	0.57	8.86	-26.22
ARS	Trailing 12	50.00	0.48	6.99	-16.34
ARS	Partial adjustment	46.03	0.38	5.76	-8.99
ARS	Lower cap	49.70	0.46	6.75	-15.50
TRY	Static 20	41.70	0.21	3.75	-1.07
TRY	Adaptive 75	53.10	0.42	7.64	-20.50
TRY	Indexed 100	56.22	0.68	10.80	-26.09
TRY	Trailing 12	48.07	0.28	4.84	-11.88
TRY	Partial adjustment	44.35	0.25	4.14	-6.07
TRY	Lower cap	47.78	0.28	4.83	-11.51

Table 15: Historical replay: adaptive rule, timely auction. The 29 overlapping windows each provide 12 months of origination and two months of possible close-out. Liquidation and credit loss use initial exposure as denominator.

Currency	Collateral	Windows	Mean liq.	Mean loss	Max loss
ARS	ETH	29	48.84	3.06	15.58
ARS	BTC	29	60.23	0.73	4.57
TRY	ETH	29	48.95	2.12	10.34
TRY	BTC	29	59.78	0.66	5.43

Table 16: ARS/ETH robustness: adaptive rule and timely auctions, 20,000 paths. Every numerical entry is a percentage of initial principal. Mixed monthly measurement uses average FX and month-end crypto; it is a diagnostic, not the main specification.

Calibration	Liquidated	Loss	CVaR ₉₉
Base: block 3	41.41	2.00	19.18
Block 1	40.24	2.20	19.01
Block 6	41.59	1.99	19.84
Omit 2020	56.38	2.48	20.30
Omit 2021	52.27	3.01	20.87
Omit 2022	11.72	0.60	11.78
Omit 2023	45.75	2.17	17.39
2020–2021	12.80	0.69	11.36
2022–2023	76.63	4.10	21.74
IID Gaussian log returns	42.11	1.58	26.00
Mixed monthly measurement	48.13	3.43	30.34

Table 17: TRY/ETH robustness, with the same definitions as the preceding table.

Calibration	Liquidated	Loss	CVaR ₉₉
Base: block 3	42.11	1.46	15.81
Block 1	41.06	1.71	15.79
Block 6	44.30	1.37	15.16
Omit 2020	56.62	1.79	16.50
Omit 2021	52.86	2.15	15.32
Omit 2022	11.53	0.44	9.41
Omit 2023	45.99	1.69	16.05
2020–2021	13.28	0.55	9.58
2022–2023	76.42	2.77	16.01
IID Gaussian log returns	43.80	1.63	26.52
Mixed monthly measurement	48.44	3.16	25.70

B Headline reconciliation

Table 18: Shared source values for abstract, narrative and generated tables: ETH, adaptive rule, timely auction. The source file is `results/main_results.csv`; exact locations and unrounded values are in `headline_reconciliation.csv`.

Currency	Metric	Raw ratio	Displayed (%)
ARS	Liquidated exposure	0.41412200	41.41
ARS	Mean credit loss	0.02002782	2.00
ARS	CVaR99	0.19182008	19.18
ARS	10% credit budget breach	0.08055000	8.05
TRY	Liquidated exposure	0.42107850	42.11
TRY	Mean credit loss	0.01462099	1.46
TRY	CVaR99	0.15807589	15.81
TRY	10% credit budget breach	0.05145000	5.15

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